



Thermodynamics II (2025,6,27)

※ Writing utensils, Function calculator, Student ID card may be brought in.

General note:

- When using a symbol that is not given in the question sentence, be sure to add its definition (meaning).
- Do not answer only the answer, but also write the equation of the derivation process.

Nomenclature (You can use following symbols without any explanation):

Q, q : heat, W, w : work, U, u : internal energy, p : pressure, T : temperature, V, v : volume, c : specific heat, H, h : enthalpy, S, s : entropy, F, f : Helmholtz free energy, G, g : Gibbs free energy, L : Latent heat, μ : chemical potential, n : molar quantity, R : gas constant

1. Answer the following questions.

- (1) Write the definition equation of Gibbs free energy and Helmholtz free energy.
- (2) Write the definition equation of the chemical potential $\mu(p, T)$ [J/mol].
- (3) Write the equation of the minute amount of Gibbs free energy $dG(p, T, n)$ in the presence of mass transfer.
- (4) There are two systems A and B which are in contact with each other. Write the conditional equation for when these two systems are in thermodynamic equilibrium.

$$\begin{aligned} dG &= d(H - TS) \\ &= d(U + pV - TS) \\ &= TdS - pdV + pdV + \mu dn - TdS - SdT \\ &= -SdT + Vdp + \mu dn \end{aligned}$$

$$\begin{cases} T_A = T_B \\ p_A = p_B \\ n_A = n_B \end{cases}$$

2. Answer the following questions.

- (1) Derive

$$dH = TdS + Vdp + \mu dn,$$

using the definition equation of Gibbs free energy and following equation

$$dG = -SdT + Vdp + \mu dn.$$

- (2) Derive three of Maxwell's relations from this equation.

$$dG = -SdT + Vdp + \mu dn.$$

- (3) Consider a non-equilibrium chemical reaction using Gibbs free energy. Explain what the Gibbs free energy is during a chemical reaction.

- (4) In situation (3), draw a graph with the vertical axis representing Gibbs free energy and the horizontal axis representing time. Enter on the graph the time at which the reaction ended, t , and the minimum value of Gibbs' free energy, $G_{\min}$.

$$dG = -SdT + Vdp + \mu dn$$

$$\begin{aligned} dG &= d(n\mu) = nd\mu + \mu dn \\ dG &= nd\mu \end{aligned}$$

$$dG = -SdT + Vdp + \mu dn$$

$$dG = nd\mu + \mu dn$$

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3. Answer the following questions about the specific heat.

(1) Using the equation (1), derive equation (2).

$$c_p = T \left(\frac{\partial s}{\partial T} \right)_p \quad (1)$$

$$\left(\frac{\partial c_p}{\partial p} \right)_T = -T \left(\frac{\partial^2 v}{\partial T^2} \right)_p \quad (2)$$

(2) When the substance is ideal gas, prove that c_p is a function of temperature only and is not affected by pressure. Note that the proof starts from equation (2).

(3) Equation (3) is Mayer's law in general expression.

$$c_p - c_v = T \left(\frac{\partial v}{\partial T} \right)_p \left(\frac{\partial p}{\partial T} \right)_v \quad (3)$$

Using equation (3), derive the Mayer's law in ideal gas.

(4) Let μ_0 be the chemical potential of 1 mol of ideal gas at temperature T_0 and pressure p_0 . find the chemical potential μ at temperature T_0 and pressure p . R R_0

4. Figure 1 shows the phase diagram of water. Answer the following questions.

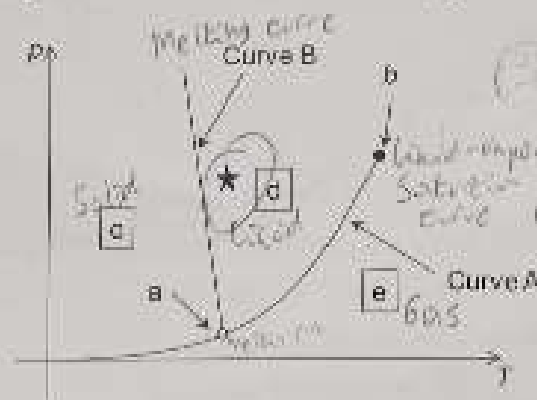


Figure 1 Phase diagram of water

(1) Answer the names of curve A and curve B.

(2) What are the phases of water in regions c, d, and e?

(3) When ice water in a glass was placed under the pressure and temperature conditions marked $\star$, the chemical potentials of each phase (μ_c and μ_d) are related: $\mu_d(T, p) < \mu_c(T, p)$.

Explain, using dG , μ_c , μ_d and dn_c , what phase will be in the glass when equilibrium is reached after a period of time. Note that pressure and temperature are constant during the process.

(4) Using volume v and entropy s of each phase c and d (v_c, v_d, s_c and s_d), show $\frac{dp}{dT}$ with Latent heat L .

in curve B. $L = T(s_d - s_c)$ $\frac{dp}{dT} = \frac{L}{T(v_d - v_c)}$ $\leftarrow \frac{dp}{dT} = \frac{s_d - s_c}{v_d - v_c}$

(5) Curve B has a negative gradient on the p - T diagram. Explain this from the Clausius-Clapeyron equation.